Software Project Management

Lecture 6

Agenda

- Software Project Management
- Roles and Responsibilities in Software Development Teams
- Stakeholders
- Scheduling
  - Gantt charts
  - Network Techniques

Software Project Management

What is a Project?

- Projects are often contrasted with process.
  - A process describes the normal day-to-day activities of an organisation, while
  - The word project is often used to describe something outside normal day-to-day work.
- A project is a unique venture with a beginning and an end, conducted by people to meet established goals within parameters of costs, schedule and quality.
  - Buchannan and Body

How do Software Projects Arise?

- Identify a gap between ‘where we are now’ and ‘what we want to achieve’ OR
- Identify a problem to be resolved OR
- Identify an opportunity for change

What is Project Management?

- Project management involves the planning, monitoring, and control of the people, process, and events that occur as software evolves from a preliminary concept to full operational deployment.
- Software project management is an essential part of software engineering.
- Why does a project need management?
  - Projects need to be managed because professional software engineering is always subject to organizational budget and schedule constraints.

Project Management

- The success criteria for project management obviously vary from project to project, but, for most projects, important goals are:
  - to deliver the software to the customer at the agreed time;
  - to keep overall costs within budget;
  - to deliver software that meets the customer's expectations;
  - to maintain a coherent and well-functioning development team.

Challenges at Managing Software Projects

- Software engineering is different from other types of engineering in a number of ways that make software management particularly challenging:
  1. The product is intangible: A manager of a shipbuilding or a civil engineering project can see the product being developed.
     - If a schedule slips, the effect on the product is visible—parts of the structure are obviously unfinished. Software is intangible. It cannot be seen or touched.
     - Software project managers cannot see progress by looking at the artifact that is being constructed. Rather, they rely on others to produce evidence that they can use to review the progress of the work.

Challenges at Managing Software Projects

2. Large software projects are often “one-off” projects: Every large software development project is unique because every environment where software is developed is, in some ways, different from all others.
   - Even managers who have a large body of previous experience may find it difficult to anticipate problems.
   - Furthermore, rapid technological changes in computers and communications can make experience obsolete. Lessons learned from previous projects may not be readily transferable to new projects.

Challenges at Managing Software Projects

3. Software processes are variable and organization-specific: The engineering process for some types of system, such as bridges and buildings, is well understood.
   - However, different companies use quite different software development processes.
   - We cannot reliably predict when a particular software process is likely to lead to development problems. This is especially true when the software project is part of a wider systems engineering project or when completely new software is being developed.

Roles and Responsibilities in Software Teams

Software Project Manager

- It is impossible to write a standard job description for a software project manager.
- Fundamental project management activities are:
  1. Project planning: project managers are responsible for planning, estimating, and scheduling project development and assigning people to tasks. They supervise the work to ensure that it is carried out to the required standards, and they monitor progress to check that the development is on time and within budget.

Software Project Manager

2. Risk management: project managers have to assess the risks that may affect a project, monitor these risks, and take action when problems arise.
3. People management: project managers are responsible for managing a team of people. They have to choose people for their team and establish ways of working that lead to effective team performance.

Software Project Manager

4. Reporting: project managers are usually responsible for reporting on the progress of a project to customers and to the managers of the company developing the software.
   - They have to be able to communicate at a range of levels, from detailed technical information to management summaries.
   - They have to write concise, coherent documents that abstract critical information from detailed project reports.
   - They must be able to present this information during progress reviews.

Software Project Manager

5. Proposal writing: the first stage in a software project may involve writing a proposal to win a contract to carry out an item of work.
   - The proposal describes the objectives of the project and how it will be carried out.
   - It usually includes cost and schedule estimates and justifies why the project contract should be awarded to a particular organization or team.
   - Proposal writing is a critical task as the survival of many software companies depends on having enough proposals accepted and contracts awarded.

People Management

Managing People

- The people working in a software organization are its greatest assets.
- It is expensive to recruit and retain good people, and it is up to software managers to ensure that the engineers working on a project are as productive as possible.
- There are four critical factors that influence the relationship between a manager and the people that he or she manages:

Managing People

- Consistency: All the people in a project team should be treated in a comparable way.
- Respect: Different people have different skills, and managers should respect these differences. All members of the team should be given an opportunity to make a contribution.
- Inclusion: People contribute effectively when they feel that others listen to them and take account of their proposals. It is important to develop a working environment where all views, even those of the least experienced staff, are considered.
- Honesty: As a manager, you should always be honest about what is going well and what is going badly in the team. You should also be honest about your level of technical knowledge and be willing to defer to staff with more knowledge when necessary.

Motivating People

- As a project manager, you need to motivate the people who work with you so that they will contribute to the best of their abilities.

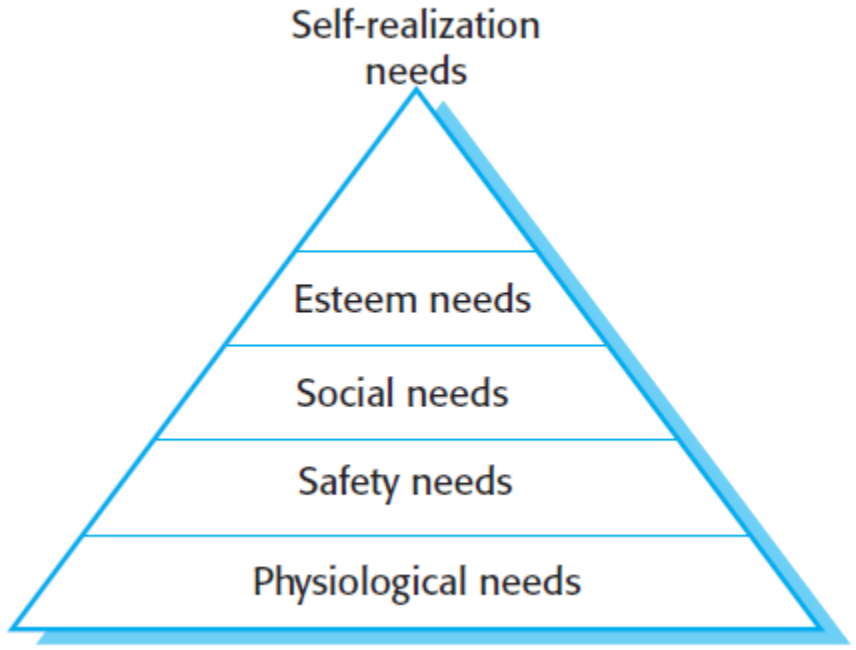


Figure 22.7 Human needs hierarchy

Other Roles

Other Roles

- Product Owner is a software development role for a person who represents the business or end-users and is responsible for working with the user group to determine what features will be in the product release.
- The Software Architect is a highly-skilled software developer that has to think through all the aspects of the project and is responsible for making high level design choices, as well as select technical standards (for instance, determines the technology stack to use).

Other Roles

- Developers are team members that apply their knowledge of engineering and programming languages in software development.
- Quality Assurance Engineers or Testers are responsible for the Quality Assurance (QA) and ensure the product meets the business requirements and that it is free of bugs, errors and defects.



https://tsh.io/blog/software-development-team-structure/
https://perfectial.com/blog/build-effective-software-development-team/
https://www.atlascode.com/blog/software-development-project-roles-and-responsibilities/

Stakeholders

What are stakeholders?

- Stakeholders are any persons (or groups) who have an interest in your project, are affected by it or who can influence its outcome.
- Stakeholders will include all the people whose work is changed in any way by a project; those who are affected by it; and all those who can in any way obstruct, block or stall the project, as well as the senior staff who grant the resources.

Categories of Stakeholders

- Senior managers who define the business issues that often have a significant influence on the project.
- Project (technical) managers who must plan, motivate, organize, and control the practitioners who do software work.
- Practitioners who deliver the technical skills that are necessary to engineer a product or application.
- Customers who specify the requirements for the software to be engineered and other stakeholders who have a peripheral interest in the outcome.
- End users who interact with the software once it is released for production use.

A Stakeholder Map

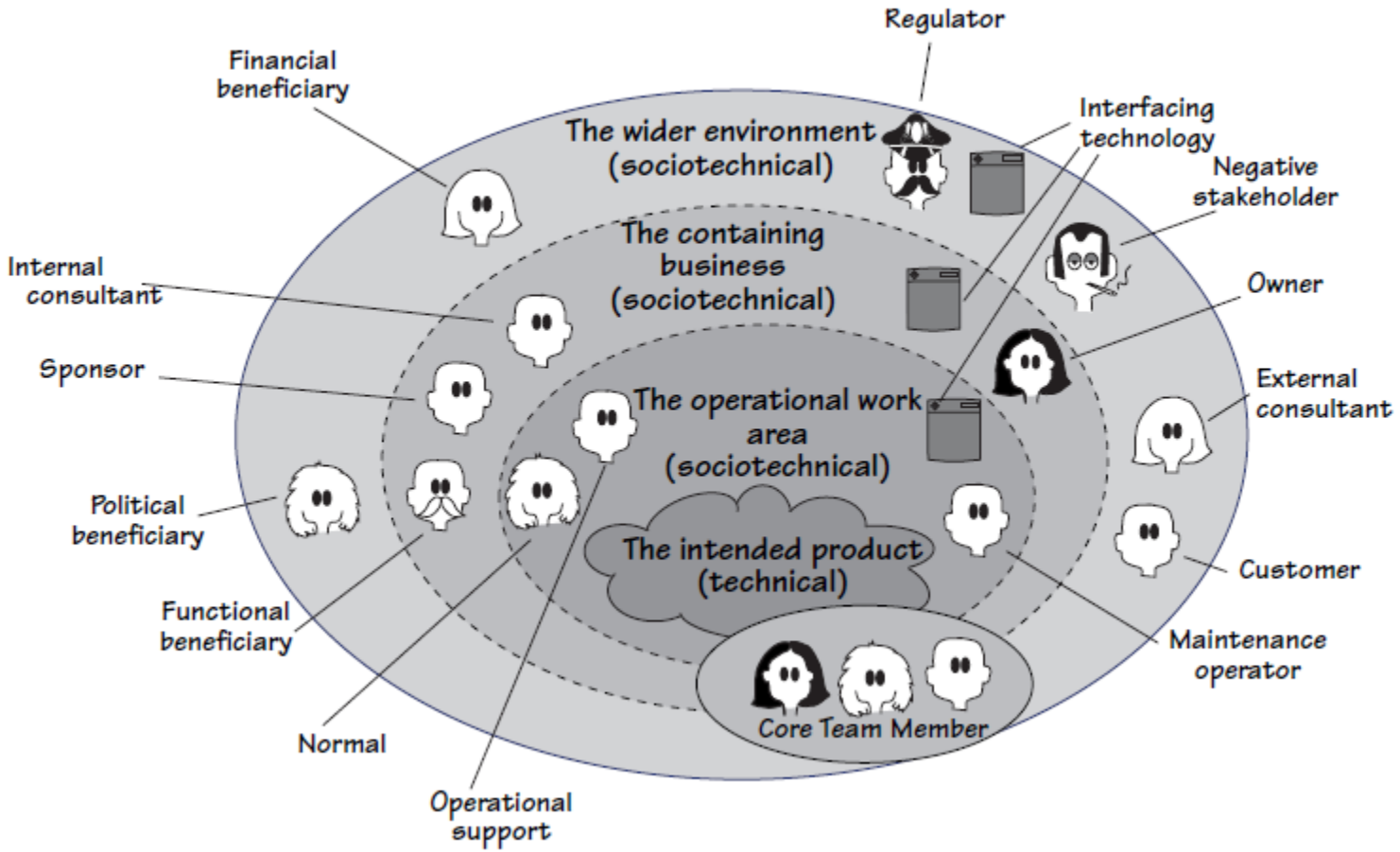
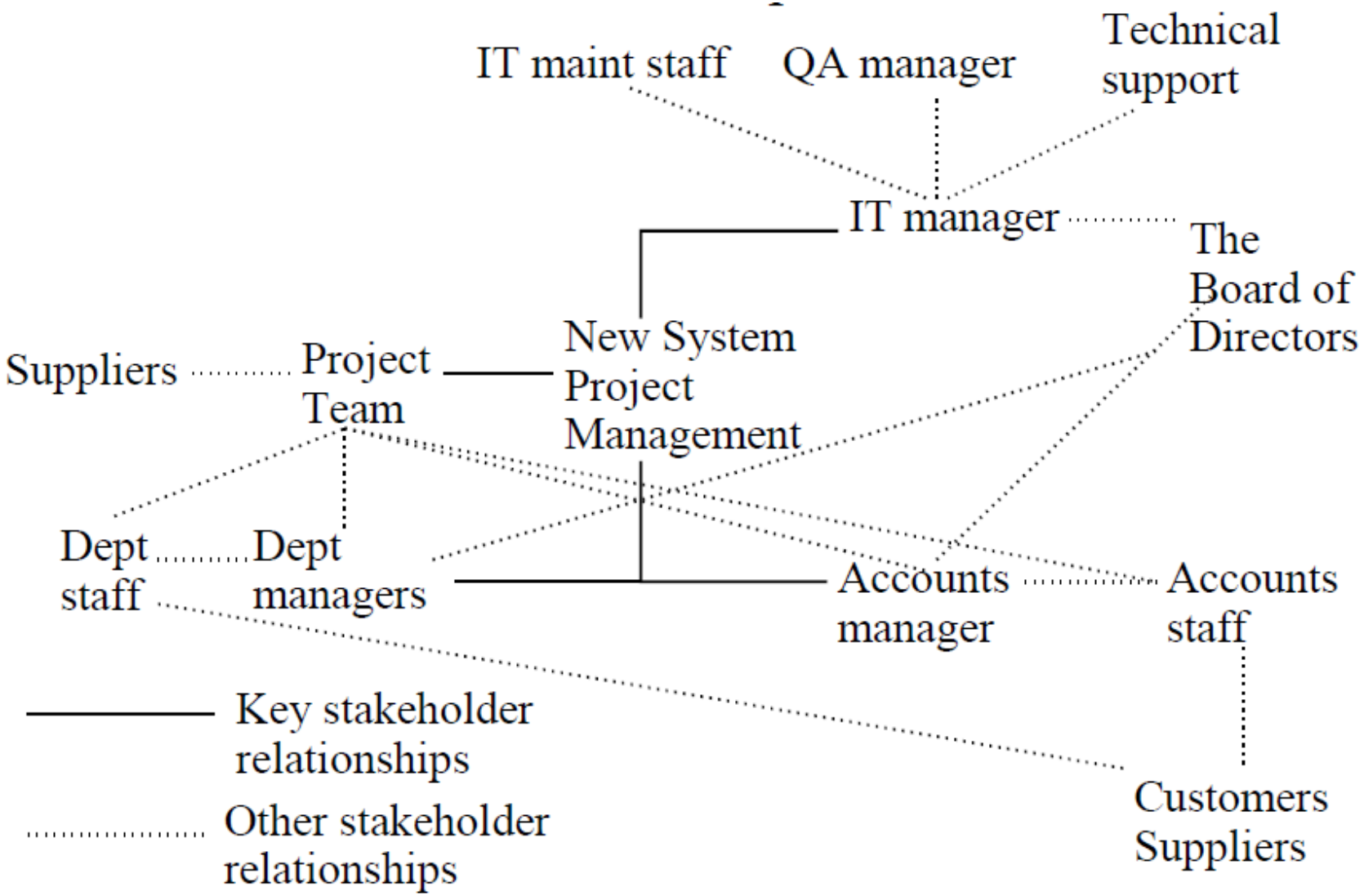


Figure 3.7

This stakeholder map shows the organizational rings surrounding the eventual product, and the classes of stakeholders who inhabit these rings. Use this map to help determine which classes of stakeholders are relevant to your project and which roles you need to represent them.

A Stakeholder Map



Scheduling

Scheduling

- Scheduling significa "*elaborar el cronograma*"
- Scheduling es una de las principales actividades de un director de proyecto.
- Muchas organizaciones de ingeniería de software desarrollan productos de alta calidad, pero son entregados muy tarde. El tiempo es dinero.
- Los proyectos con retraso son caros para cliente, el contratista o ambos.

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Scheduling

- Para intentar asegurar que un proyecto sea terminado a tiempo es necesario preparar un cronograma detallado de cada actividad en el proyecto y monitorear.
- Elaborar y monitorear un cronograma detallado no es suficiente para garantizar que un proyecto será terminado a tiempo.

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Fases y milestones

Fases y milestones

- Un proyecto puede ser dividido en **fases**. Por ejemplo, instalar un nuevo equipo en una fábrica podría tener las siguientes fases: diseño, compra, instalación y puesta en marcha.
- Los eventos importantes o principales **milestones** en el proyecto son los puntos de terminación de cada fase.
- Dividir un proyecto en fases secuenciales a menudo no es posible: usualmente hay un número de actividades que se traslapan en el tiempo.

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Milestones

- Siempre se debe identificar un número de eventos importantes claramente definidos en un proyecto, los cuales pueden ser utilizados como milestones.
- Una característica de un buen milestone es que no puede existir duda alguna acerca de si ha sucedido o no. Por ejemplo, durante la construcción de un buque podría ser el lanzamiento del buque al agua.

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Milestones

- En el caso de un milestone para la terminación de una fase de diseño por ejemplo, se debe definir claramente los productos que debe ser entregados para poder decir que se ha completado el diseño.
- Un buen milestone también debe ser significativo.
- El término **baseline** es utilizado para representar el estado de un proyecto o alguna parte identificable de un proyecto en un instante en particular. Se utiliza usualmente para el estado del final de una fase significativa del trabajo.

Baseline

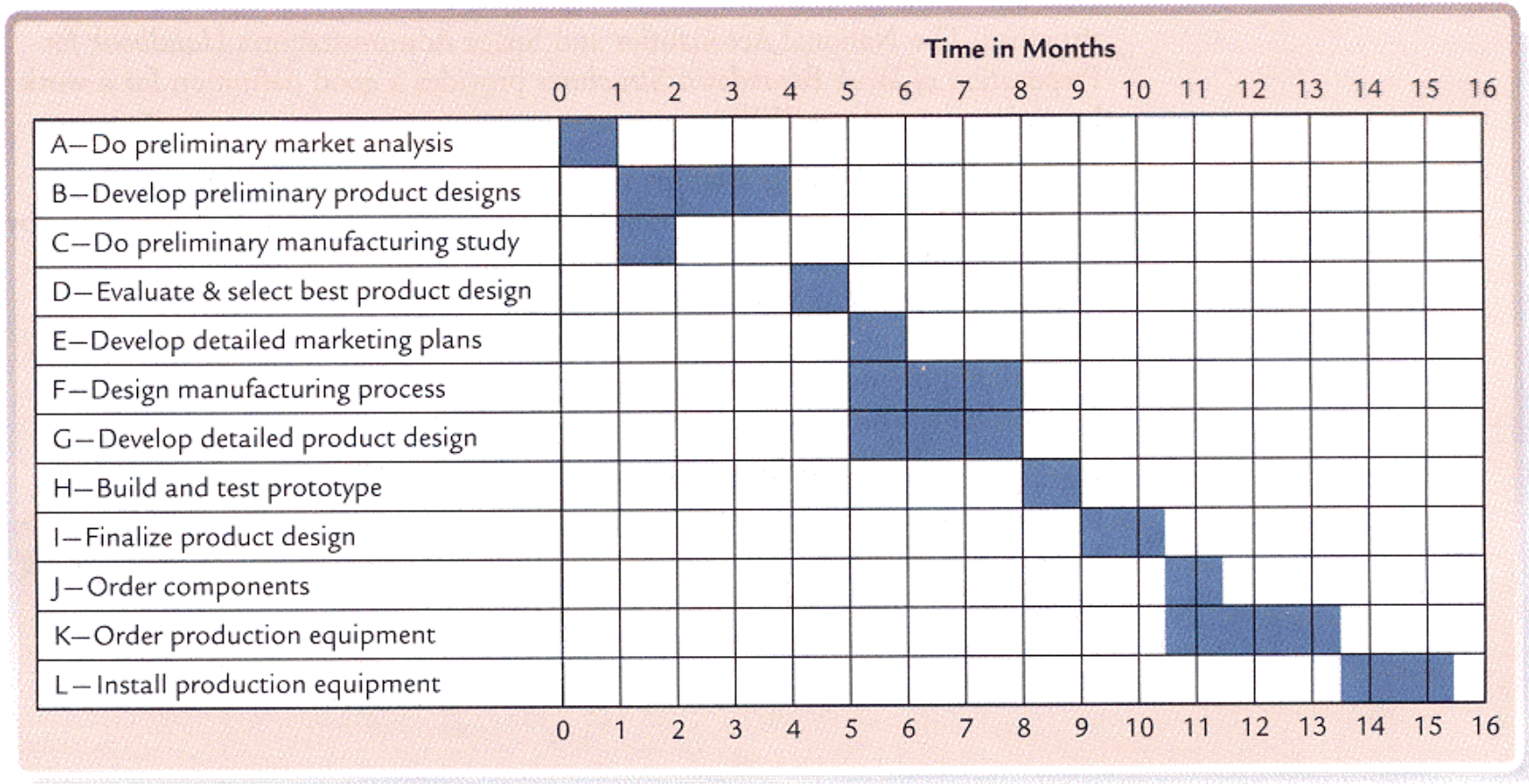
- La siguiente fase se construye sobre el baseline alcanzado al final de la fase anterior.
- Por ejemplo, un posible baseline es que el diseño de un grupo de ítems ha sido terminado y aprobado. Usualmente, se planificaría que este baseline sea alcanzado antes que la construcción de esos ítems pueda empezar.

Gantt charts

Gantt chart

- La idea de un cuadro de Gantt es listar todas las actividades de un lado de la página y utilizando una escala de tiempo horizontal, dibujar una barra para cada actividad con la finalidad de representar el periodo sobre el cual debe ser desarrollada.

Gantt chart



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Gantt chart

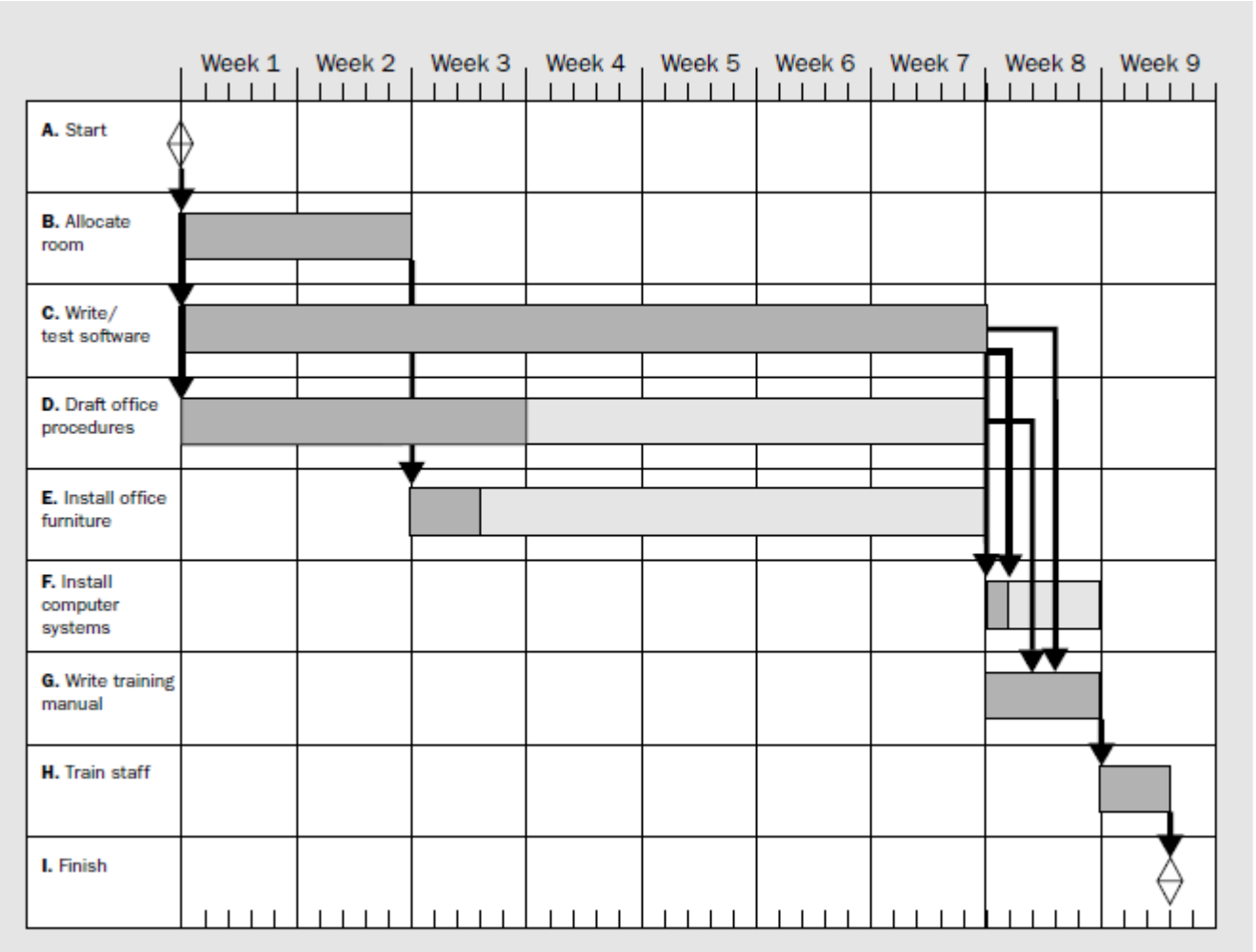


Fig. 2.13
Gantt chart

Gantt chart

- Existen variaciones del cuadro de Gantt los cuales incluyen mayor cantidad de información.
- Es fácil de entender, no obstante el contenido del mensaje en un simple cuadro de Gantt es bastante limitado.
- Entre las variaciones, se utiliza el color o un sombreado para representar las actividades que son responsabilidad de una persona o departamento, o para indicar el avance de cada actividad.

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Gantt chart

- Un aporte significativo es resaltar la cadena de actividades secuenciales que determinan el tiempo mínimo requerido para el proyecto. Se dice que estas actividades están en el **critical path** (camino crítico). Demoras en cualquiera de estas actividades incrementaría la duración del proyecto.

Limitaciones del Gantt chart

- No es fácil mostrar las dependencias de una actividad con otra, excepto por inferencia.
- No se puede apreciar el camino crítico del proyecto.

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Network Techniques: activity-on-arrow networks

Network Techniques

- Existen paquetes informáticos para producir cronogramas basados en técnicas de redes (ej.: Lucidchart, EdrawMax).
- Existen algunas variantes de las técnicas de redes: **critical path scheduling,** critical path analysis y PERT (Project Evaluation and Review Technique).
- La idea principal de la red es representar gráficamente la relación secuencial entre las actividades que van a ser desarrolladas en el proyecto.

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Network Techniques

- En **contraste** al cuadro de Gantt en donde dibujar las barras no puede ser hecho hasta que las fechas o al menos el orden relativo y las duraciones sean fijas, en las redes el objetivo es desarrollar un diagrama **sin conocer al inicio la fecha real** en la cual las actividades van a ser desarrolladas.
- El diagrama de red de las actividades y sus dependencias entre sí, no cambian significativamente aún cuando la duración estimada para las actividades cambia o alguna nueva actividad es añadida.

Network Techniques

- Para desarrollar una red, el planificador debe responder dos preguntas para cada actividad:
  - ¿Qué se debe hacer antes de iniciar esta actividad?
  - ¿Qué se puede hacer una vez que esta actividad es terminada?
- Una vez que las cadenas de actividades dependientes ha sido identificada, una tercera pregunta puede ser realizada acerca de cada actividad:
  - ¿Qué actividades pueden ser realizadas al mismo tiempo que esta actividad?

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Activity-on-arrow networks

- Cada actividad es representada por una flecha que inicia y finaliza en un nodo (dibujado como un círculo).
- Cada nodo representa un **evento**, un punto con tiempo de duración igual a cero que significa la terminación de todas las actividades dirigidas a ese nodo.
- Los números debajo de cada flecha representan la duración de cada actividad.

Activity-on-arrow networks

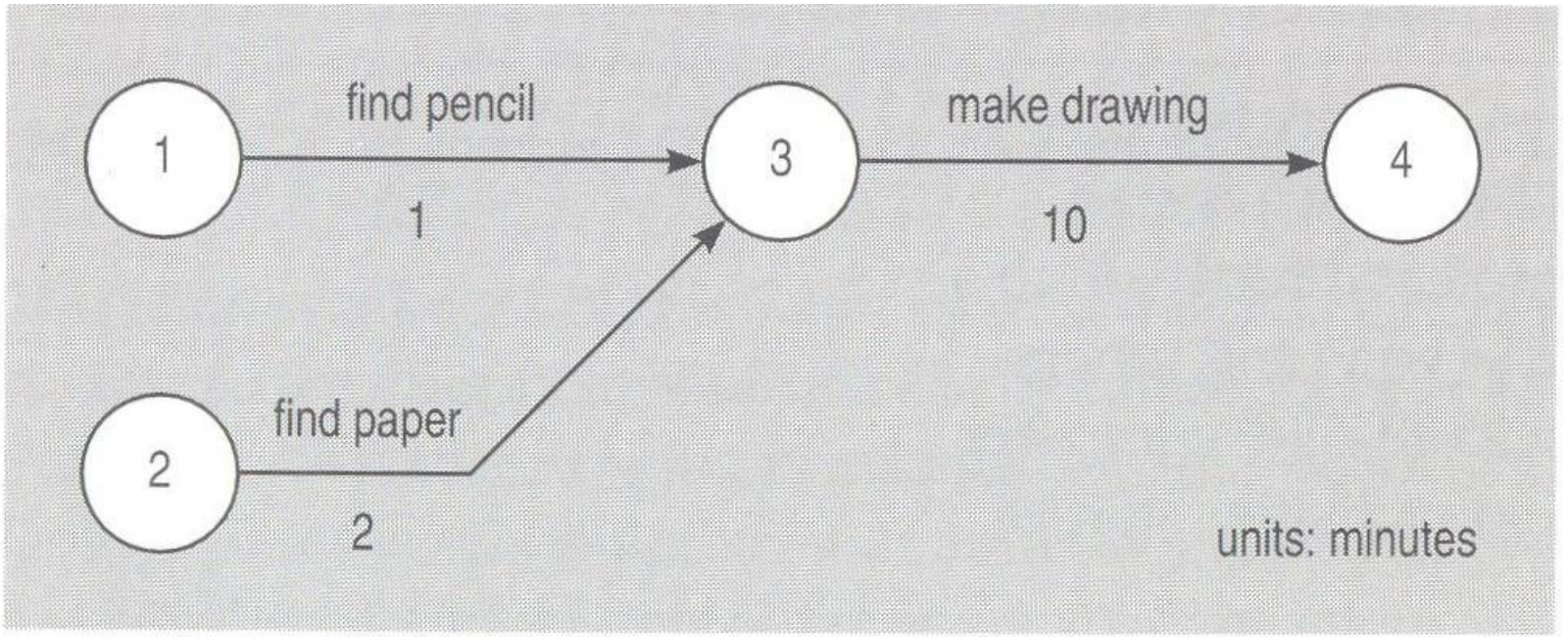


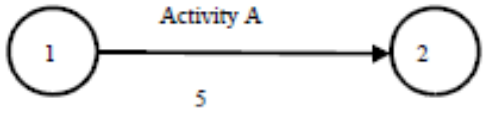
FIGURE 3.8 An activity-on-arrow network

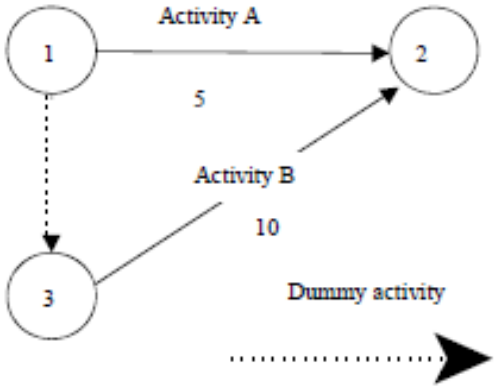
- Los números en los círculos representan eventos

Dummy activities

- En virtud que los caminos a través de la red son identificados por eventos únicos, solo una actividad puede unir un par de nodos.
- Las actividades dummy introducen un nodo adicional, exclusivamente para efectos de identificación.

Dummy activities





Dummy activities

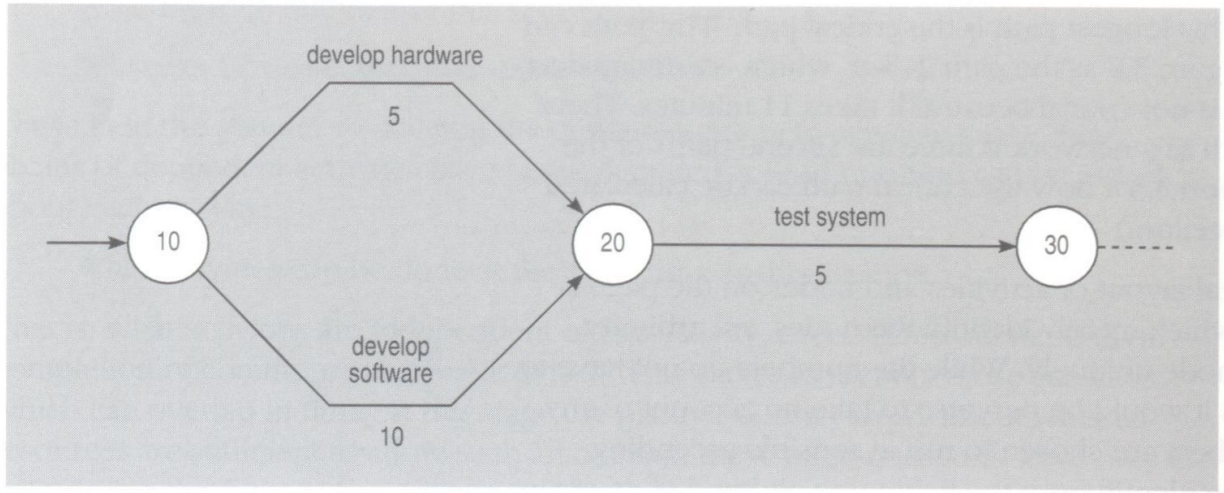


FIGURE 3.9 Two activities with the same identification

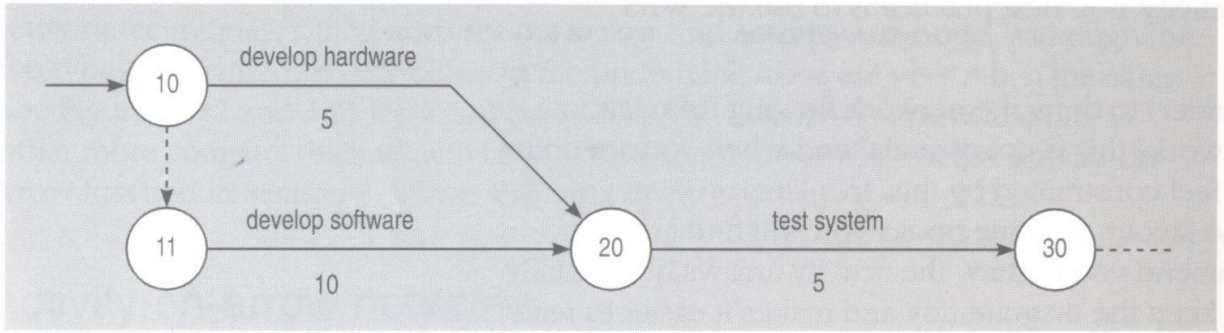


FIGURE 3.10 The project network of Figure 3.9 with a dummy activity inserted

Dummy activities can be used whenever you wish to show a dependency without a physical activity. The use of dummies will also sometimes improve clarity.

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Inicios múltiples y fin

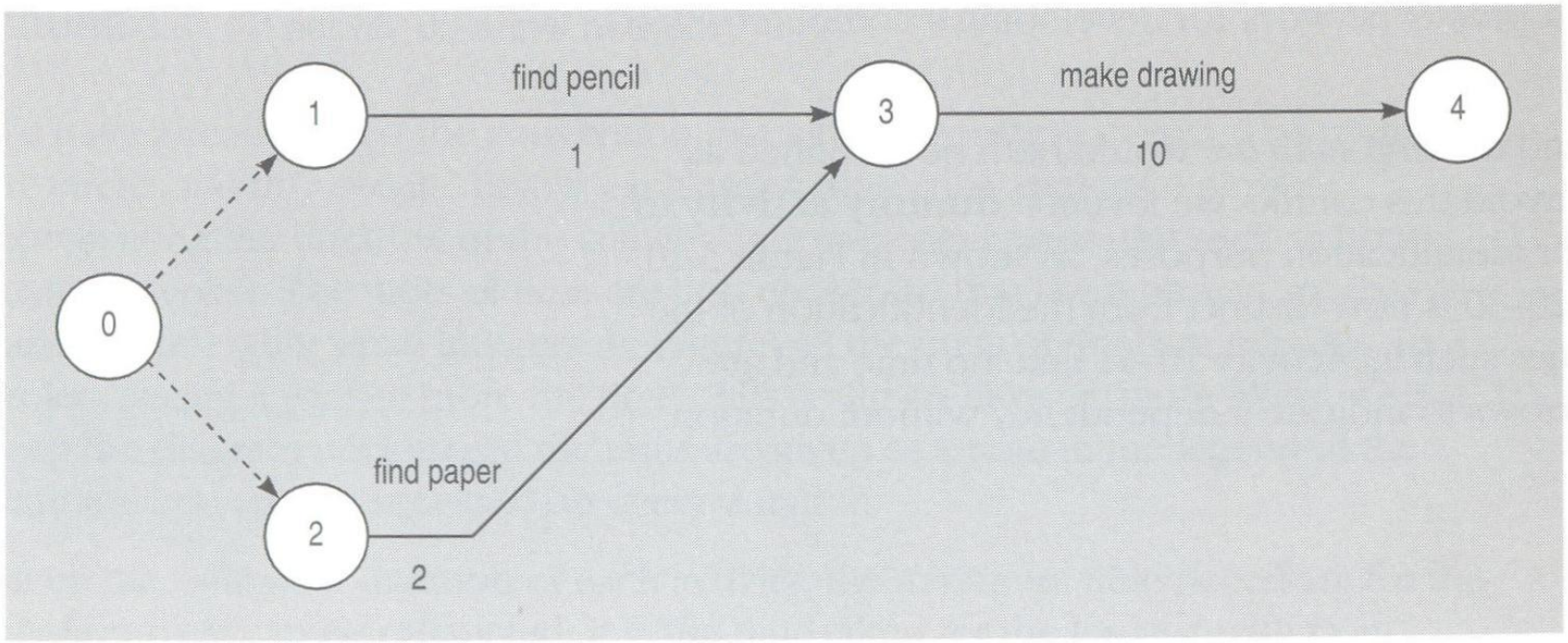


FIGURE 3.11 A single starting node

Activity-on-arrow networks

- Las redes **NO** pueden contener lazos

Loops

Project networks are not allowed to contain loops, a cycle of activities where each one requires the completion of its predecessor. This rule prevents us from representing redesign of a component as shown in Figure 3.12.

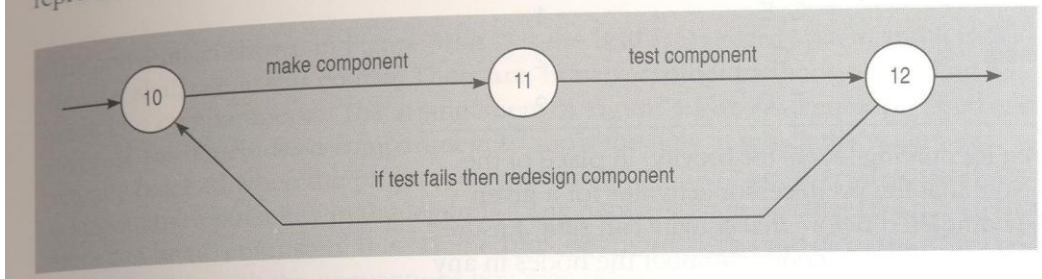


FIGURE 3.13 An inadmissible loop

The reason for the rule is that such a network cannot be analysed to show how long it will take. The existence of a loop highlights to the project planner that this project might go on forever. To avoid this catastrophe, some time limit has to be set on the internal iterations so that they can be represented by the more acceptable Figure 3.13.

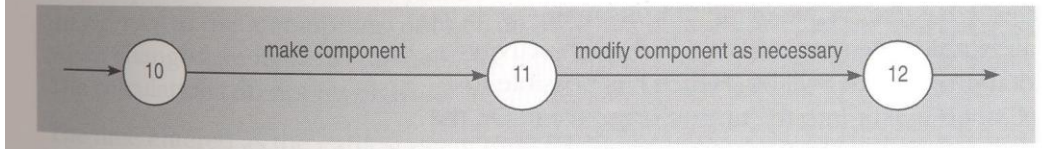


FIGURE 3.13 An acceptable finite repetition

Next On

Take Away Points

- What is a project?
- Why does a software project need to be managed?
- Which are the main responsibilities of a software project manager?
- Roles in software development teams
- What is a stakeholder?
- Scheduling concepts
- Gannt charts vs. Activity-on-arrow networks

Further Reading

- Field and Keller, “Project Management”
  - Chapter 1: Project Initiation
  - Chapter 3: Planning
  - Chapter 4: Building and Leading the Team
- Pressman and Maxim, “Software Engineering”
  - Chapter 31: Project Management Concepts
- Ian Somerville, “Software Engineering”
  - Chapter 22: Project Management
- Robertson and Robertson: Mastering the Requirements Process
  - Chapter 3: Scoping the Business Problem

Further Reading

- Project Manager, “5 Stakeholder Map Examples With Analysis”
  - https://www.projectmanager.com/blog/stakeholder-map-examples
- Garla S. Naven, “Stakeholder Mapping and Goal Alignment for Software Architects”
  - https://medium.com/@naveen.garla/stakeholder-mapping-and-goal-alignment-for-software-architects-a2f8876622d2
- Simply Stakeholders, “Stakeholder Mapping: Methods, Benefits & Examples”
  - https://simplystakeholders.com/stakeholder-mapping/
- Mural, “How to create a stakeholder map [templates & examples]”
  - https://www.mural.co/blog/stakeholder-mapping

Further Reading

- NOAA, “Working with People: Stakeholder Analysis Exercise”
  - https://coast.noaa.gov/data/digitalcoast/pdf/stakeholder-analysis-worksheet.pdf
- UNIR, “Stakeholders mapping: concepto, usos y cómo desarrollarlo”
  - https://www.unir.net/revista/empresa/stakeholders-map/
- Miro, “Stakeholder Mapping”
  - https://miro.com/blog/stakeholder-mapping/
- Canva, “Creating a stakeholder map: How-tos, templates, and tips”
  - https://www.canva.com/online-whiteboard/stakeholder-maps/
- Lucid, “A quick and dirty guide to stakeholder mapping”
  - https://lucid.co/blog/a-guide-to-stakeholder-mapping

Next Lecture

- Resource Allocation and Scheduling in Software Projects
- Software Size and Estimation